



Faclon Labs Data and AI Infrastructure

Decision Intelligence at scale to unlock Latent Potential of each Plant

Improve Production
Throughput and Yield

Reduce Operational
Cost and Wastage

AI Agents Enable Decision
Intelligence at Scale

www.faclon.com



Faclon Labs at a Glance

Leaders in Industrial Data Integration, Visualization, and AI based Optimization



50+
Customers

200+
Deployments

60k+
Assets
Connected

5B+
Data Points
Daily

120+
Employees



Rishi Sharma
CEO
B.Tech IIT Bombay



Utkarsh Srivastava
CTO
Ex-White Hat Hacker



Parth Chaudhary
President
B. Tech IIT Bombay



Vivek Singh
Chief Data Scientist
B.Tech IIT Kanpur



Archit Naraniwal
Chief Sales Officer
B. Tech IIT Bombay



Mukesh Kumar
India Operations
B. Tech IIT Bombay

Backed By



Paula Mariwala
Founder, Stanford Angels



Sarjan Shah
MD, Group Satellite



Jay Kotak
Co-head, Kotak 811



Siddharth Mehra
ED, Savita Oil

Key Customers

Trusted and Ideal digital partner for Industrial setups, commercial setups and OEMs

Automobile & ancillary



TATA HITACHI



Consumer Goods



Metals



HINDUSTAN ZINC



Cement



Pharma



Consumer Durables



LUMINOUS

Hamilton



Commercial Infra



Chemicals



Energy, Oil & Gas



OEM tie-ups



Partnerships



What we do, How we Deliver

Powerful Ecosystem of platform offerings for rapid digital transformation

Critical Outcomes



Automating Decisions

- Diagnostic Analytics
- Prescriptive Analytics
- Optimal Control Strategy
- Digital Twin Simulations



Problem Visibility

- KPIs Benchmarking
- Dashboards, Alerts & Reports
- AI based Detection Models
- Golden Batch Analytics



Data Unification

- Integration with all IT and OT data sources
- Workflow Digitization
- System of Records
- Data Contextualization



The Operating System for Physical Infrastructure

Hyper Customised Factory Specific Intelligence at Scale

01 Production
Throughput & Quality

03 Maintenance
Availability & Maint. Cost

02 Energy & Utilities
Equip. Efficiency & Specific Consumption

04 Safety
Compliance & Incidents



Connecting People, Assets, Process and Data at Scale

Ideal Mix of Capabilities to Achieve Transformational Results

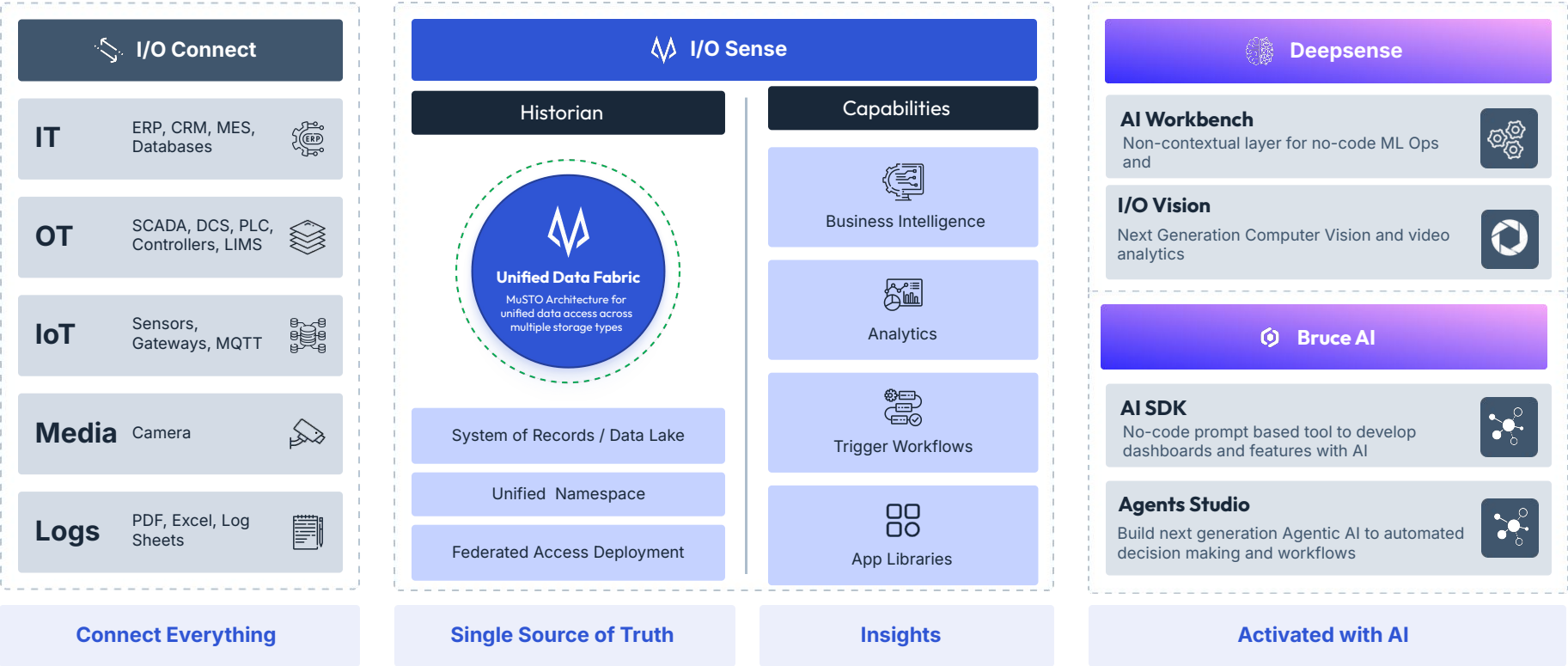
Ideal digital transformation product to measure, find issues and improve productivity



Platform Architecture Overview

Powerful Ecosystem of platform offerings for rapid digital transformation

CxO Suite | Digital COE | Plant Heads | Plant Managers



Siloed Data to Insight: The Digital Transformation

Integration with all data to create a Single Source of Truth

KPI Benchmarking & Drill-Down Analysis: Making Performance Visible

Machine learning models to generate powerful insights

GenAI as drive digital plant co pilot and automation



I/O Connect

IT-OT Integration

OT Integration

Smart server and client middleware to enable integration with PLCs, DCS, Historians, etc. using protocols like Modbus, Profinet, OPC UA/DA, IEC 61850, IEC 104, Ethernet IP, MQTT, SNMP, etc.

Local Historian

Historian module at the 3.5 layer to store data for months to years and securely sent to on-prem or cloud IT platforms

API Gateway

For API integration with third party IT systems with ERP, MES, CMMS etc.



I/O Sense

Data Visualisation

Use Case Specific Tools

Specific, prebuilt and ready to go tools and dashboards for more than 30+ use cases

End to End Configurable Tools

Smart set of configuration tools to store data, contextualize data, on board new data pipelines, create dashboards, reports, alerts, workflows, maps, APIs, Business logics etc.

Role Based Access

Strong admin panel to onboard data streams, create users and manage deployments & VMs. Mangle access provided to different users



DeepSense

Machine Learning

Deploy ML codes with AI Workbench

Provides a comprehensive AI no code toolkit to build, train, validate, and deploy ML models like forecasting, predictions, optimization etc. using no-code AI Workbench.

I/O Vision for Computer Vision

Leverages I/O Vision to analyse the camera data to perform different computer vision use cases like PPE detection, conveyor defect detection. Power full platform to optimally run on edge compute systems.



Bruce AI

Gen AI

No code features with AI SDK

Prompt and build the tool, feature, dashboard of your own choice within hours. Complete integrated with the platform

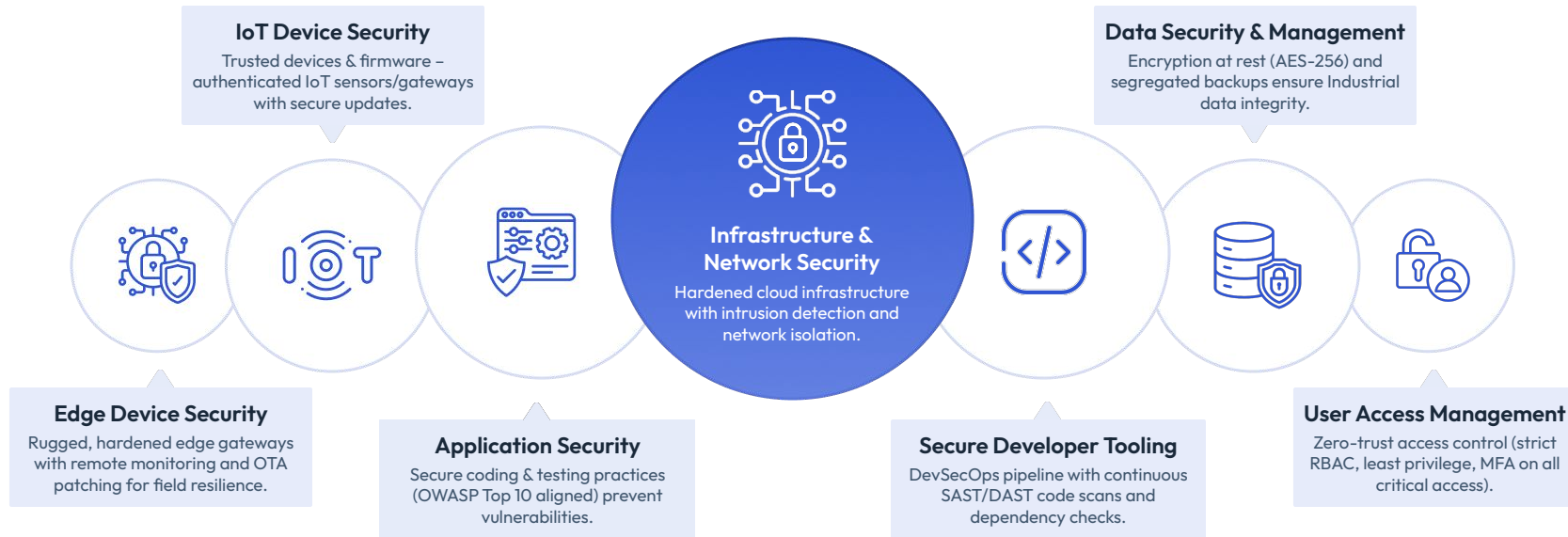
Automate with Agent Studio

Build your own AI agents by creating the workflows in agent studio. Manage MCPs, knowledge graphs, vector DBs, Ontology. Build Agent Workflows. Choose LLM model of your own choice. Build tools to do automated RCA, recommendations, copilots etc.

From Connected Data to Intelligent Decisions

Security Stack Overview

Faclon Labs Multi-Layered Security Stack at every touchpoint



Transport Layer Security

End-to-end encrypted data transit (TLS/IPsec tunnels safeguarding all communications)



Continuous Monitoring & Response

24/7 threat monitoring (WAF, IDS) and incident response plans ready



Governance & Compliance

Security policies, audits and SOC 2-aligned processes ensure trust and regulatory compliance

Digital Use Cases

One-platform, unlimited applications

Production



OEE &
Downtime
Monitoring



IT-OT Integration
and Data Lake



Digital Logbook
& Workflow
Automation



AI Driven
Parameter
Optimization



Digital Twin



Product
Traceability



LIMS & Quality
Management



Vision - Surface
Defect
Detection



Statistical
Process Control



Golden Batch
Analysis

Energy & Utilities



Energy
Management



Utilities
Efficiency
Management



Water
Management



Intelligent HVAC
Automation



Solar and Wind
Monitoring



SEC Root Cause
Analysis Agent



Energy Demand
Forecasting



Power Plant AI
for NHR
Reduction

Safety



Fire Hydrant
Monitoring



Vision - PPE &
Road Safety

Maintenance



Steam Trap
Failure Detection



Vibration
Monitoring
(CBM)



VFD Health
Monitoring



Busbar / Bus Duct
Temperature
Monitoring



Vision - Foreign
Particles



UPS Health
Monitoring



Advance
Maintenance
Management



Vision - Conveyor
Defect Detection



Vision - Particle
Size Analysis



Relay &
Transformer
Monitoring



Power Quality
Monitoring



Anomaly
Detection

Clients	Context	Project Description
	Model Digital Plant	- Deployed a Flagship Model Digital Plant project for integrated cement plant of JSW wherein we integrated with all the DCS, energy meters, vibration sensors, solar inverters, ERP and lab equipments to deliver a single source of truth, data lake and taylor made dashboards for different departments. - Implemented AI based use cases like cement strength prediction, cyclone health prediction, Kiln fuel ratio optimization to increase AFR, RCA and recommendation for high specific energy consumption in mills and kilns.
	Energy Management and Reliability	- Saved 2+ Cr annually by implementing online steam trap failure detection for 250 HP steam traps across 2 plants - Demonstrated 10+ Cr potential annual saving by digitizing 2500+ meters across 4 plants with deviation analysis at all the levels of energy distribution. Saved 6 critical motor failures with alerts based on motor load. - Implemented 100+ vibration sensors for critical motors and gearboxes to reduce unplanned downtimes and maintenance cost. - Implemented health monitoring of VFD drives to increase their life and reduce breakdowns.
	IT-OT Integration	- Data from various 18 OT sources such as PLCs, SCADA systems, SQL databases, and Excel files are collected (4000 tags) and consolidated into a central server. This ensures a unified data environment for analysis and reporting. - The management and plant teams utilize KPI dashboards—covering station-wise cycle time, energy consumption, pipe production, and machine utilization—to gain real-time visibility. This has enabled quicker response to downtimes, leading to reduced production losses and improved overall plant productivity, energy efficiency, and quality.
	Machine Performance Analytics	Integrated with 30 machine PLCs including Extruder, Braider, etc. and implemented tracking of important KPIs like OEE, Availability, Powerformance, quality, production, downtime reasons, line speed, thickness, Cpk, Ppk, process parameters etc. - Integrated with ERP to map the data with production order for WIP traceability purpose. The success of this implementation has led to scale up across 400+ machines.

Clients	Context	Project Description
	HVAC Optimization, Steam trap & Solar Monitoring	- Implemented IoT based fully automated system for Chiller plant, AHU and Packaged units at 2 locations - Implemented real time steam trap failure detection for 100 HP and LP steam traps - Implemented centralized solar efficiency monitoring of 43 sites for tracking energy efficiency
	Predictive Maintenance	Detected 15+ critical failures at early stage for JSW Energy. Deployed 1000+ vibration sensors across 5 plants. Reduced unplanned downtimes, revenue loss and maintenance cost.
	Vision AI for Safety	Implemented IO Vision for compliance of Plant Safety for ISO 39001:2012 Road Traffic Safety (RTS) Management Systems and ISO 45001 Occupational Health and Safety Management System standards for 196 camera for 6 use cases including PPE + Hard hat Detection, Fall + Loitering, Entry - Exit Monitoring, Intrusion Detection, Speed Monitoring
	AI-Driven Boiler-Turbine Performance Optimization	- Developed AI-driven models to improve net heat rate, emissions & availability for a 21 MW cogeneration power plant covering all the assets i.e. boiler, turbine, generator, and auxiliary systems. - Integrated with the plant DCS & lab data and deployed AI models for optimal combustion, soot blowing schedule, aux power and steam reduction, reduced attemperation, condenser - cooling tower circuit, LOI reduction etc.
	Smart Maintenance	CMMS-ERP integrated workflows enabling automated work orders, technician assignment, material planning, ERP PR & Reservation creation, man-hour logging, and real-time closure for preventive and corrective maintenance. AI-driven condition monitoring using vibration, partial discharge sensor, and fire hydrant monitoring to predict failures, trigger corrective actions, estimate remaining useful life (RUL)
	IT-OT Integration	Implemented on-premise centralized monitoring by Integrated with all the machines DCS and energy meters across the plant for real time monitoring, data historian, reporting and alerting.

Clients	Context	Project Description
	LIMS Platform	Implementing comprehensive and integrated LIMS system integrating 10+ plant sections enabling real time monitoring, process adjustments, capability analytics, traceability, faster RCA, automated reporting, alerts and capability analysis. It is integrated with lab machines, ERP, and MES.
	Energy and Utilities Management	- We deployed a IO Sense Utility Management System for ITC starting with the Trichy plant, covering 250+ assets across power, water, boiler, HDS and LPG systems. - Expanded the deployment to Uluberia plant with 100+ energy meters integrated into the platform. - Then, Implemented at Mysore for 150+ assets including energy meters, flow meters, chillers, air compressors and other utilities. With three plants successfully operational, Now scaling to 7 more plants, with 2 plants already in the implementation pipeline
	Energy Management	Deployed power analytics and energy management for 140+ Energy Meters for first plant, 256 energy meters for the second plant. With successful deployment, now scaling to 3 more plants.
	Bombay Airport Digitization	Monitor and Control 75 AHUs at the domestic terminal to maintain temperature and improve energy efficiency. - Implemented wireless and battery based - ambient temperature & humidity across 300 locations to ensure consistent and comfortable cooling throughout the terminal. - Installed UPS health monitoring system for 200 UPS to ensure reliability of power backup and proactive issue detection.
	Adani Natural Resources	Implemented a pilot for real time monitoring of exploration Drill Rig Performance for Adani Natural Resources
	Cold Storage Monitoring	Implementing cold storage temperature monitoring for Flipkart's 1000 dark store locations across India.
	Steam Trap Monitoring	Delivered a successful pilot for 30 qty steam traps, Implementation ongoing for over 900 qty steam traps for Bathinda site.
	Cell House Monitoring	- Deployed an integrated monitoring system using RFID-based cathode traceability and load cell measurements at the stripping stage to track zinc yield per batch, monitor cell efficiency, and assess cathode health in real time. - Real-time cell temperature and voltage monitoring at 108 cell locations to proactively monitor zinc harvest cycle
	Vision AI for Asset health	Monitoring the health of 3 conveyor belts (Shuttle conveyor - proximity switch fault detection, tray conveyor- alignment, wheel wobbling, misalignment etc. and belt conveyor - spillage, wear & tear etc.) with vision camera

Clients	Context	Project Description
	OEE / Production Excellence	- Driving Operational Excellence across the 7-step bottle manufacturing line via real-time OEE monitoring and Andon-integrated downtime classification (Prod/Maint/Qual), leveraging automated analytics to eliminate shopfloor bottlenecks. - Optimizing MTO agility through assembly-level planning and dynamic ETC(Expected time to complete) tracking, synchronizing rated cycle times with planned downtimes for precise schedule adherence across the entire line.
	Process Excellence	Paste Mixers: Deployed a Diagnostic Process Twin to digitize mixing phases, benchmark cycle times, and correlate dynamic process parameters for pinpointing bottlenecks and driving predictive quality.
	Quality Traceability	Implemented an end-to-end digital thread for battery testing—mapping QR-scanned SKUs to HRD recipes and SQL databases to ensure fail-safe label printing only for quality-passed units.
	Process Excellence	- Digitized end-to-end boom manufacturing workflows replacing manual checksheets with real-time production, quality, and compliance tracking. - Enabled complete operator productivity visibility with automated downtime capture, driving data-backed decisions across fabrication shop operations. Established full traceability with first pass yield tracking, failure analytics, and ISO-compliant digital document control system
	Process Excellence	- Integrated 14 environmental chamber machines with real-time monitoring of temperature, humidity, and pressure across product testing workflows. - Digitized end-to-end test planning, execution, and approval workflows with multi-level validation and operator-driven test logging. Enabled Energy optimised test scheduling and ERRC-based recommendations using historical test data to optimize utilization and reduce defects.
	SOP Adherence	Enforced Computer Vision based Process SOP adherence with PLC interlocks on a 15-step carburetor assembly line—enabling real-time step validation, preventing sequence deviations, and reducing rejections by ~50%.
	OEE	Transformed 20+ injection moulding machines of godrej's vendors with real-time OEE, downtime, cycle time analytics, mold recommendation engine.

We're here as your

Digital Backbone to Drive Operational Efficiency and Decision Intelligence

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